

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

COMPUTER COMMUNICATIONS AND NETWORKS

Friday 6th May 2011

14:30 to 16:30

Year 3 Courses

Convener: K. Kalorkoti
External Examiners: K. Eder, A. Frisch

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

1. (a) Define “protocol.” [2 marks]
- (b) Give two advantages of protocol layering. [2 marks]
- (c) Give two key differences between the Internet (TCP/IP) and OSI network architectures. [2 marks]
- (d) What is the difference between network architecture and an application architecture? Give three examples of application architectures. [4 marks]
- (e) Describe two uses of Domain Name System (DNS) other than hostname to IP address translation and host/mailserver aliasing. [3 marks]
- (f) Address Resolution Protocol (ARP) and Domain Name System (DNS) both depend on caches; ARP cache entry lifetimes are typically 10 minutes, while DNS cache is on the order of days. Justify this difference. What undesirable consequences might there be in having too long a DNS cache entry lifetime? [6 marks]
- (g) Consider the effect of using slow-start on a link with a 10ms round-trip time and no congestion. The receive window is 24KB and the maximum segment size is 2KB. How long does it take before the first full window can be sent in one transmission round? [2 marks]
- (h) Consider sending a large file from one host to another over a TCP connection that has no loss. Suppose TCP uses AIMD for its congestion control *without* slow start. Assume *CongWin* increases by 1 MSS (maximum segment size) every time a batch of ACKs is received. Also assume approximately constant RTTs (round-trip times).
 - i. How long does it take for *CongWin* to increase from 1 MSS to 7 MSS (assuming no loss events)? [3 marks]
 - ii. What is the average throughput (in terms of MSS and RTT) for this connection over that period (from sub-part (i))? [1 mark]

2. (a) List five key differences between IPv4 and IPv6. [5 marks]
- (b) Discuss two mechanisms for transition from IPv4 to IPv6. [4 marks]
- (c) What is the counting-to-infinity problem? [1 mark]
- (d) Describe a general mechanism to resolve the counting-to-infinity problem. [4 marks]
- (e) Route oscillations occur in (vanilla) link-state and distance-vector routing algorithms when the link cost metric is congestion dependent. Describe an approach to avoid such oscillations while still using congestion based link cost metric. [4 marks]
- (f) Suppose a router has the following CIDR entries in its routing table:
- | Net/Prefix | Next Hop |
|-------------------|-----------------|
| 128.96.39.0/24 | Interface 0 |
| 128.96.39.128/25 | Interface 1 |
| 128.96.40.0/25 | Router 2 |
| 192.4.153.0/26 | Router 3 |
| default | Router 4 |
- For each of the following IP addresses, what does the router do if a packet with that address arrives? (i) 128.96.39.10; (ii) 128.96.40.12; (iii) 128.96.39.151; (iv) 192.4.153.17; (v) 192.4.153.90. [5 marks]
- (g) As each network interface at a router or host is assigned a unique IP address, why is a separate MAC address (also called physical address) needed? [2 marks]

3. (a) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [7 marks]
- (b) Describe self-learning in a switch. [2 marks]
- (c) How does self-learning help in the case of a host moving between two access points, both connected to a common switch via two different ports at the switch? [2 marks]
- (d) Suppose the data portion (including the headers) of a packet contains a sequence of 8 bytes, each corresponding to 8-bit unsigned binary representation of the following integers: 243, 81, 128, 64, 32, 16, 27, 9. The corresponding sequence of hex values for those integers is: 0xf3, 0x51, 0x80, 0x40, 0x20, 0x10, 0x1b, 0x09. Compute the Internet checksum for this data. [4 marks]
- (e) Consider the 4-bit generator, $G = 1001$, and suppose that the data portion of a bit stream to be transmitted prior to Cyclic Redundancy Check (CRC) calculation is 11001001. Show the actual bit stream transmitted. Suppose the leftmost bit in the transmitted bit stream is inverted due to noise on the transmission link. Show that this error is detected at the receiver's end. [6 marks]
- (f) Give two reasons why networks might use an error-correcting code instead of error detection and retransmission. [4 marks]